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Generative Adversarial Network as Data Balance and Augmentation Tool in Histopathology of Breast Cancer

Marya Ryspayeva
Department of Computer Engineering
Astana IT University
Astana, Kazakhstan
m.ryspayeva@astanait.edu.kz
marya.rys1@mail.ru

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***Abstract*—This paper explores Wasserstein Generative Adversarial Network Gradient-Penalty (WGAN-GP) for data balance in the medical domain where data scarcity and imbalance are common. The study applies Transfer Learning with pre-trained models from ImageNet on histopathological breast cancer data, both unbalanced and balanced. WGAN-GP was used to overcome the challenge of generating synthetic images to balance the data and improve the accuracy of the classification task. The highest results were shown by VGG16 with a balanced dataset by WGAN-GP in 300 epochs (95.40% accuracy, 96.56% sensitivity, 94.91% specificity). Results showed an improvement in accuracy from 84.25% to 95.40%.**

Keywords—generative adversarial network; GAN, WGAN-GP; data balance; histopathology; breast cancer; transfer learning; artificial intelligence

I. INTRODUCTION

Today, data plays an essential and valuable role in different aspects of our lives. However, raw data cannot be used directly for analysis in any field, and it has to be processed to extract useful information from it. For instance, receiving brain MRI images, radiologists need to read and analyze them to conclude if there is any abnormality or not. Scientists create different systems to process data and methods of AI (Artificial intelligence) comes into play [1]. AI methods have occupied niche in different medical domains research. They can solve the tasks of segmentation, classification, object detection, and recognition and predict the probability of diseases, lesions, locations, and other targets with human-level accuracy. Scientists deploy developed algorithms to help improve the detection and diagnosis of such diseases as COVID-19 [2], breast cancer [3], liver cancer [4], brain tissues segmentation [5], and other abnormalities and diseases.

Most of the papers overview and integrate the Deep Learning (DL) methods into the algorithms, which gives 85%-99% of accuracy. DL is the most appropriate AI method to classify and recognize medical imaging tasks [6]. In turn, developers leverage Deep Convolutional Neural Network (DCNN) as the algorithm reaches high accuracy. Recent studies show that researchers apply the AlexNet, ImageNet, and Visual Geometric Group (VGG) architectures of DL to the developed software. However, training DL properly in the medical domain is a challenging task that requires large datasets to receive high results. DL needs at least ten thousand images; even if they are, the problem of labeling also exists [7]. Researchers face the problem that labeling the images and receiving the Ground Truths (GTs) is a highly time-consuming and expensive task, especially in the medical domain [8]. For example, some public breast cancer datasets are used to train AI and presented in scientific articles. However, they are characterized by a small number of samples and an imbalance in the classes. Moreover, the deficit of the data provokes overfitting of the model of the networks, which leads to poor generalization [9].

Unequal distribution of the classes in the dataset yields biases in one of the classes, which reflects in the results. DL allows boosting the number of the samples by data augmentation applying different transformations such as rotation, flipping, shearing, translation, cropping, and others. However, another tool can generate synthetic data based on the samples, generative adversarial networks (GANs) [10].

The applications of GANs are diverse. However, author decided to focus on the problem of data augmentation and class balance. GAN is a potential solution for the annotation of medical images [11]. Furthermore, one of the main problems with DL neural networks is that training a model from scratch requests many data, and even if such data are available, it will take a long time. It is proposed to generate data by GAN to balance classes in the dataset and apply Transfer Learning (TL), which uses pre-trained models to overcome these challenges with data balance and training time. TL has been gaining popularity for its easiness and rapidity to

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train datasets in recent years. The facility concludes by transmitting received knowledge from one trained model to another [12].

The paper is organized as follows: "Introduction" describes the challenges of the data to train AI in the medical domain. 'Related work' analyzes state-of-the-art papers applying GAN as data balance and augmentation. In the section "Methodology," it is described the proposed methods' details. "Experiments and Results" presents and analyzes the results, followed by "Discussion" and "Conclusion" with an inference on issues and future work.

II. Related Work

The GAN network consists of two networks of a generator and a discriminator. The first one tries to fool the second one by creating fake images, while the discriminator tries to distinguish between real and fake images of the generator. As the discriminator gets better at discriminating, the generator also gets better at generating. Eventually, the discriminator can be excluded and only use the generator to produce as many images as possible to augment the data. For instance, "Pix2Pix" conditional GAN is forced to produce particular kinds of images. It is necessary to have pairs of images such that if the first image of 'domain A' is passed to the GAN, it can produce its counterpart of 'domain B' [13]. The generator is fed an image as an input; then, the discriminator is fed the same image as an input when it tries to distinguish between fake and real. Another example of GAN is "CycleGAN," where there is no need to have paired images, which is difficult to obtain in the medical domain. Instead, it cleverly bypasses this by using cycle consistency, a straightforward idea. Instead of using one GAN, a CycleGAN uses two GANs. It uses the first GAN to go from the first domain to the second and then uses this output as an input to the second GAN to reconstruct the input of the first GAN [14].

Abdul Waheed et al. [2] proposed to use Auxiliary Classifier GAN (ACGAN) to generate synthetic images of chest X-rays, which allowed to get a significant amount of data. The data shortage was observed due to the rapid spread of the pandemic and the short time to collect large datasets of the X-rays. ACGAN improved the accuracy of the detection of the infection from 85% to 95%, which are encouraging results. Authors claimed that the classical data augmentation tool in the DL frameworks is not enough to generate unseen data, while GAN overcomes this limitation. The dataset comprises only ~400 images with infection and ~700 normal images. Furthermore, it is necessary to consider that this number of images was divided into training, validation, and test subsets to decrease the number of samples during the training. It used 2000 epochs to train the proposed GAN for 5 hours, and as a result, ~1400 normal and ~1700 positive images were generated. Precision and recall also were increased without and with synthetic samples from 89 to 0.96 and from 0.69 to 0.90, respectively [2]. ACGAN relates to the conditional GAN where the discriminator predicts class labels.

Maayan Frid-Adar et al. [4] faced the problem of a lack of large datasets. The size of the computer-tomography images was 182 liver lesions of three classes. More samples need expert knowledge and time resources to annotate and get GTs. Firstly, the authors applied classical data augmentation by rotating, translating, and scaling to increase the dataset size. Shearing and elastic deformations were not used to keep liver lesions characteristics. After that, augmented data was fed to the Deep Convolutional GAN (DCGAN) to generate three classes of images. As in [2], the GAN consists of two networks of discriminator and generator, where the last one tries to create real images, and the first network to distinguish them as fakes. As a result, 18,000 synthetic images helped to improve specificity and sensitivity from 78.6% to 88.4% and 85.7% to 92.5 respectively.

Faezehsadat Shahidi et al. [15] developed a system that generates high-resolution images from low resolution to reconstruct breast cancer histopathology by GAN. Generator network combines residual blocks and self-attention layers, improving image quality. The dataset combines Breast Cancer Histopathological Image Classification (BreakHis) and Cancer Metastases in Lymph Nodes (Camelyon). The first includes ~8000 images with different resolutions, with ~2500 and 5500 benign and malignant samples. Moreover, the second one consists of 400 high-resolution images. The authors leveraged wide-attention Super-Resolution GAN with Wasserstein Gradient penalty (GP) to stabilize the model in the training step. Around 300 epochs were trained and gave 99.49% of accuracy in high-resolution images.

Mai Bui Huynh Thuy and Vinh Truong Hoang [16] reported that using a GAN helped improve the accuracy in the classification of the histopathology of breast cancer significantly. They highlighted that training a GAN is challenging in the medical domain. The synthetic images were generated by StyleGAN and Pix2Pix GAN, classified by VGG16 and VGG-19 models of TL. The combination of GANs gave them ~8000 synthetic images and increased accuracy from 91.7% to 95%. However, the authors mentioned that the developed GAN generated some trainable features instead of the same images as the originals.

Tasleem Kausar et al. [17] proposed an algorithm of stain acclimation generative adversarial network (SA-GAN), consisting of one generator and two discriminators. The discriminators take into account the color and texture features of the images. The authors combined three datasets of the histopathological images of breast and colon cancers of ~2500 images. The proposed algorithm can classify images with an accuracy of 93%, which is higher than 6% compared to other reviewed methods.

Asha Das et al. [18] indicated a problem with the required number of images in the datasets and their GTs due to time demands and the high cost of the data labeling. They developed a new algorithm of NAS-SGAN (Nuclear atypia scoring), where the discriminator is trained in unsupervised and supervised ways. The novel algorithm could achieve remarkable accuracy of 98%, with only 20% presented GTs.

As can be seen [2, 4, 15, 16, 17, 18, 19], most of the datasets are presented as imbalanced, which negatively influences the accuracy. GAN is used as one of the solutions to generate minority classes, and the abovementioned papers showed a significant improvement of 7-10% in the results. GAN learns distinctive features and applies these features in

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generating new synthetic data. Then synthetic and real data are fed to the pre-trained models of TL to classify images. Many authors leverage TL because of the scarcity of data. If author generates many images with GAN, there is a problem with the overfitting.

III. MATERIALS

This paper used a dataset of Invasive Ductal Carcinoma (IDC), a type of breast cancer. The dataset was taken from the Kaggle website https://www.kaggle.com/datasets/paultimothymooney/breast-histopathology-images. It is a challenging task to detect the disease accurately. The images are presented as histopathological with the regions of the IDC cells. Dataset consists of 162 high-resolution images divided into patches of 50 x 50 pixels of RGB 3-channels. The number of patches is 277,524, with 198,738 negative and 78,786 positive samples [20, 21].

IV. METHODOLOGY

A. WGAN-GP as Data Balance and Augmentation Tool

One of the main challenging tasks in training GAN is failing in convergence and generating poor images. WGAN-GP can overcome these obstacles with regular training and better performance than other GANs [23]. Usually, GAN has two adversarial networks of generator and discriminator, where the first one creates noise distribution as input while the second network identifies input as fake.

Most GANs are presented as a minimax game where discriminator minimizes values and divergence between data distributions. However, it often leads to vanishing gradients and discriminator saturation that complicates the training process.

A classical GAN is shown as a minimax game, where the generator aims to minimize the loss function while the discriminator tries to maximize it. Equation 1 shows this process in a mathematical way [10]:

$$\min_{G}\max_{D} \times \mathbb{E}_{x\sim p_r}[\log D(x)] + \mathbb{E}_{\tilde{x}\sim p_g}[\log(1 - D(\tilde{x}))] \quad (1)$$

where G - generator, D - discriminator, x - real data sampled from data distribution p_r, $\tilde{x}$ - fake data generated by generator from random noise distribution p_g, $\mathbb{E}_{x\sim p_r}[\log D(x)]$ - the expected value of the discriminator's output when it is given a real sample x from the training data; $\mathbb{E}_{\tilde{x}\sim p_g}[\log(1 - D(\tilde{x}))]$ is the expected value of the discriminator's output when it is given a generated sample from the generator.

Equation 1 presents the Jensen-Shannon divergence between real and synthetic data. In WGAN, equation 1 is replaced with the Wasserstein Distance function that calculates the distance between points if they are moved from one distribution probability to another (Equation 2). Additionally, the discriminator is a critic predicting the Wasserstein distance by optimizing weights [22].

$$\min_{G}\max_{D\in\mathcal{D}} \times \mathbb{E}_{x\sim p_r}[D(x)] + \mathbb{E}_{\tilde{x}\sim p_g}[(D(\tilde{x}))] \quad (2)$$

Equation 2 presents 1-Lipschitz functions of Lipschitz constraints where weight clipping enforces Lipschitz from explosion or vanishing because weights are pushed of the critic to the two extreme clipping values. Furthermore, the weights clipping threshold has to be fine-tuned [23]. However, WGAN produced such famous problems of GANs as vanishing gradients, slow convergence, and unstable training [24].

Gradient Penalty was integrated into the WGAN algorithm instead of weight clipping to overcome the abovementioned problems. (Equation 3) [23]:

$$\min_{G}\max_{D\in\mathcal{D}} \times \mathbb{E}_{\tilde{x}\sim p_g}[(D(\tilde{x}))] - \mathbb{E}_{x\sim p_r}[D(x)] + \lambda\mathbb{E}_{\hat{x}\sim p_g}[(\|\nabla_{\hat{x}}D(\hat{x})\|_2 - 1)^2] \quad (3)$$

where λ - the penalty coefficient, $\mathbb{E}_{\tilde{x}\sim p_g}[(D(\tilde{x}))] - \mathbb{E}_{x\sim p_r}[D(x)]$ - the Wasserstein distance between the real data distribution and the generated data distribution, $(\|\nabla_{\hat{x}}D(\hat{x})\|_2 - 1)$ - the L2 norm of the gradient of the discriminator with respect to the distribution of the real data.

In WGAN-GP, the loss function is penalized if the gradient moves to the maximum values that keep from exploding and vanishing gradients. Leveraging Gradient Penalty allows not to use batch normalization and fine-tuning of hyperparameters. As a result, WGAN-GP trains the networks with high stability, optimization, and fast convergence than classical GAN and WGAN [23].

Fig. 1 shows the used architecture of WGAN-GP, which consists of two networks of generator and critic which is called instead of the discriminator. Histopathological breast cancer acts as input to the generator. In its turn, the critic tries to distinguish between real and fake images.

A latent noise vector is presented as a random normal distribution with a 0.02 standard distribution and a latent dimension of 128. The first layer is the dense layer, where the input of 64x64x3 pixels is reshaped to the 1-D flatten layer with the activation function of Leaky ReLU to output an 8x8x512 tensor. Generated tensors are concatenated and passed through four Conv2DTranspose layers (transpose convolutional), then upsampled from 8x8 to 64x64 with filters of 4x4 and strides 2x2. After each convolutional layer, batch normalization layers with Leaky ReLU are followed except for the last layer, where the hyperbolic tangent (Tanh) activation function is used. The total number of the parameters is ~16 million. As a result, the network generates images with the output of 64x64x3. Fig. 1 shows the architecture of the generator. The critic model is a network with two outputs of 0 and 1 (fake and real). The model consists of three convolutional layers with downsampling from 64 to 8 pixels with the same kernel, stride, and ~400 thousand of parameters. The architecture of the critic is shown in Fig. 1.

B. Transfer Learning for Classification of Histopathological Breast Cancer

Researcher proposes exploiting TL by fine-tuning a fully-connected layer of the convolutional networks. The DCNN method is based on pre-trained models on ImageNet Data set, comprising more than 1,000 categories. In this case, the ImageNet dataset acts as initialization to extract features from the large to the required small dataset, reaching high effectiveness.

Training the model from scratch is a time- and resource-consuming challenge. As a result, TL is more suitable to predict a target in the medical domain. As pre-

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trained models, VGG-16, VGG-19, Xception, InceptionResNetV2, Inception-V4 are used. The comparison between unbalanced and balanced datasets is presented in the "Experiments and Results" section.

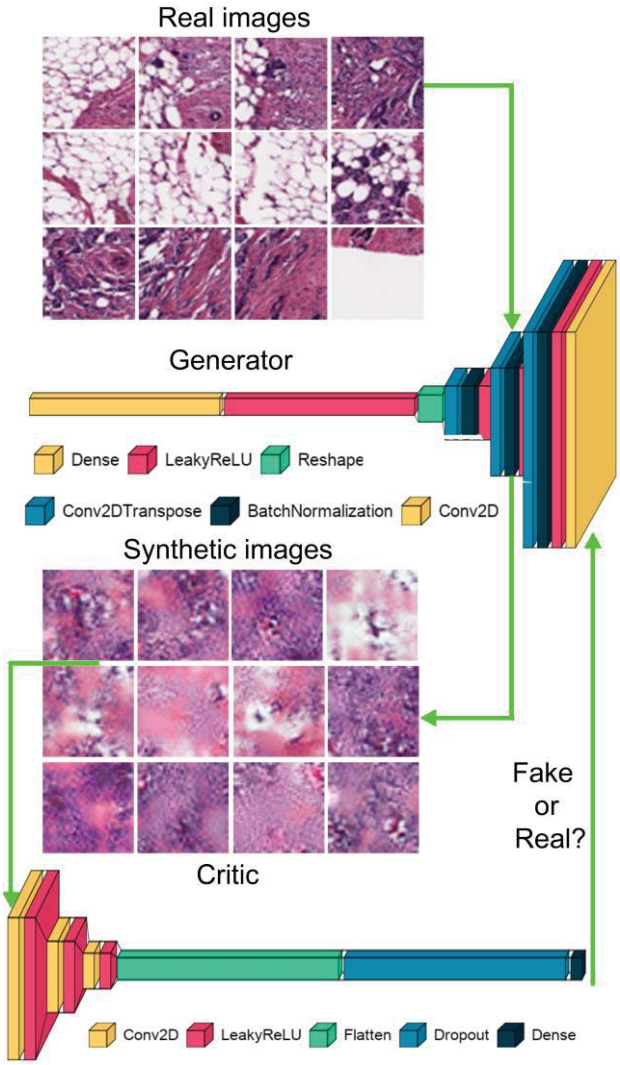

Figure 1. WGAN-GP architecture with Generator and Critic networks

Pre-trained models improve the accuracy of breast classification by CNN-depth in convolutional layers stacked on top of each other. The strategy is to freeze all layers until the second fully-connected layer, while it will be fine-tuned. Fine-tuning of the second fully-connected layer assists in the classification of non-IDC and IDC cancer (Fig. 1). Modifications are suggested in the second fully-connected layer by its fine-tuning. The Tanh function substitutes the traditional activation function ReLU. Xavier weights nominate all inputs for uniform distribution of them.

V. Experiments and Results

In this section, the experiments and results are presented. The section consists of two parts: i) results of training WGAN-GP and generating synthetic results and ii) Transfer Learning results to classify unbalanced and balanced datasets by data augmentation and balanced data with synthetic samples.

A. *WGAN-GP Results*

GAN is trained in different epochs: 300, 500, and 800. with a learning rate=0.0001, Adam optimizer, batch size=128. Elapsed time: 300 epochs = 14.3 hours, 500 epochs = 20.8 hours and 800 epochs = 23.4 hours. Fig.2 shows the training process GAN and convergence of Discriminator and Generator Losses. All figures (Fig. 2 a, b, c) demonstrate fast convergence in 150-200 epochs which indicates the characteristics of WGAN-GP.

As a result, the total number of the generated images made up ~84,000 of the positive histopathological breast cancer images, which allowed us to balance data with the majority of the class. Fig.2, d shows the synthetic samples in three different variants of epochs. At first glance, the images do not differ much, perhaps only in sharpness. However, the classification results show a clear difference in learning.

In this paper, it was applied random flipping of horizontal and vertical, zooming with a 0.2 range, shearing with 0.2 ratios, and rescaling between [0, 1]. Random contrast leads to misclassification and class confusion, which is not considered as a type of transformation.

B. *Histopathological Breast Cancer Classification Results*

Five pre-trained models with a different configuration of the dataset were trained: i) unbalanced dataset, ii) balanced dataset augmenting data by affine transformations iii) balanced dataset by GAN with three sets of epochs (300, 500, 800). The hyperparameters for training are learning rate = 0.0001, Adam optimizer, batch size = 32, patience = 10 epochs, dropout rate = 0.5, loss = “categorical cross entropy”

The results of training unbalanced dataset display that VGG16_Unbal classified images with the lowest accuracy / sensitivity / specificity of 86.30% / 91.03% / 76.04% among VGG16 variants (Table 1). Data augmentation balance in VGG16 improved the metrics slightly, just to 1-2%.

The highest results were shown by VGG16 with the balanced dataset by WGAN-GP in 300 epochs and amounted to the accuracy/sensitivity/specificity of 95.40% / 96.56% / 94.91%. Other epochs of balancing by WGAN-GP resulted in lower accuracy of 3-4% due to the overfitting problem. In Fig.2, d, it is seen that training more epochs in GAN does not improve the quality of the synthetic images significantly. Three hundred epochs can be enough to recognize features in the images and classify samples with reasonable accuracy, sensitivity, and specificity.

Furthermore, VGG19 and Xception balanced with GAN of 300, and 500 epochs have comparable numbers, which differ only 1-1.5%. VGG19_Bal_WGAN-GP-500 can classify non-IDC and IDC histopathological images with 94.33% accuracy, whereas correctly detected samples make up 95.74%. Additionally, Xception showed the results almost equal without and with balancing with two methods.

Along with VGG16_Bal_WGAN-GP-300, InceptionV4 (InceptionV4_Bal_WGAN-GP-300) shows one of the highest results of accuracy / sensitivity / specificity of 95.36% / 96.82% / 93.69%.

Pre-trained models VGG16 and InceptionV4 gave the highest result, over 95% of accuracy, two of them with VGG19 indicated a sensitivity above 95% and specificity of around 93%. While training unbalanced dataset can detect cancer with an average accuracy of 87%.

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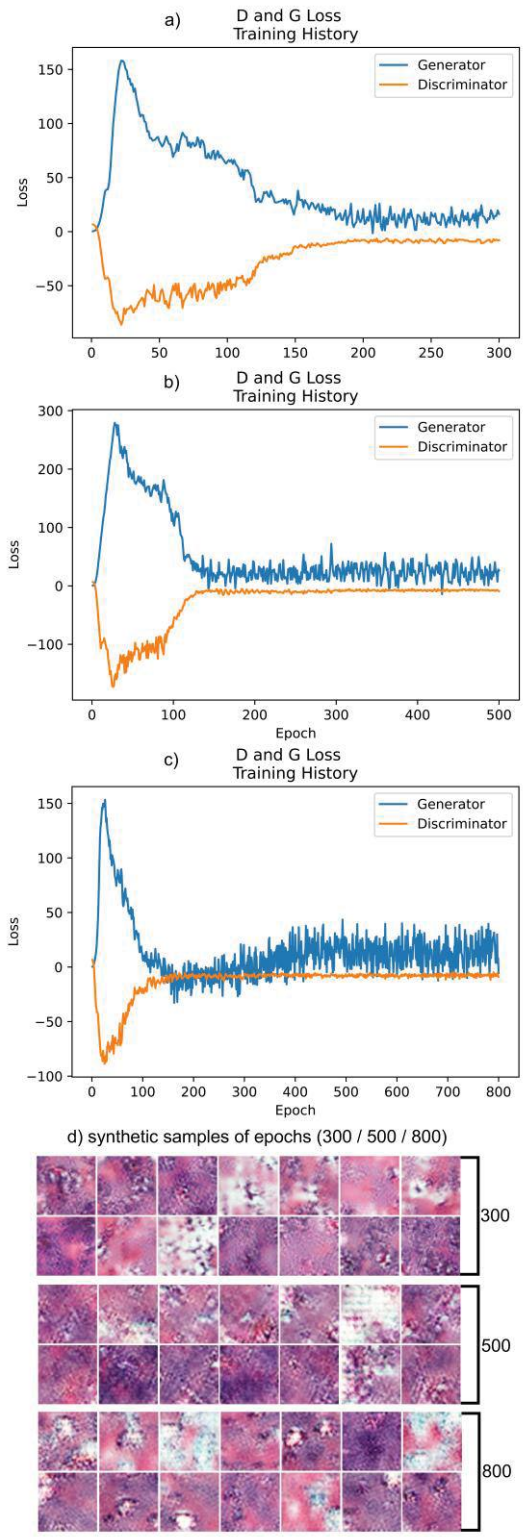

Figure 2. WGAN-GP architecture with Generator and Critic networks

In conclusion, the accuracy was enhanced from 5-10% y balancing with GAN. However, developers have to onsider different parameters and also complexity with time.

VI. Discussion

Many researchers apply DL approaches and receive ıe highest accuracy and robust classification in the ıedical domain [25-30]. They face the problem of a mited number of data, time-consuming data annotation, nbalance and overfitting problems. Researchers leverage ifferent standard techniques to data augment, such as data ugmentation of DL frameworks and programming ınguage methods (oversampling, undersampling, ssigning weights). However, applying standard data alance methods cannot improve the results dramatically s GAN. Today, the diversity of GANs allows one to hoose the appropriate network for a different task. This aper investigated WGAN-GP as a data balance and ugmentation tool and developed a system to classify istopathological breast cancer images. According to the eceived results, the research has promising results to llow us to say that GAN can be used as a balance and ata augmentation tool.

TABLE I. Classification Results of the Experiments

Method	Accuracy	Sensitivity	Specificity
VGG16_Unbal	86.30	91.03	76.04
VGG16_Bal_DataAug	87.61	93.80	89.18
VGG16_Bal_WGAN-GP-300	**95.40**	**96.56**	**94.91**
VGG16_Bal_WGAN-GP-500	92.20	96.36	93.55
VGG16_Bal_WGAN-GP-800	94.19	93.89	90.84
VGG19_Unbal	85.22	90.31	76.1
VGG19_Bal_DataAug	89.56	91.58	85.16
VGG19_Bal_WGAN-GP-300	93.27	94.88	90.36
VGG19_Bal_WGAN-GP-500	**94.33**	**95.74**	**93.56**
VGG19_Bal_WGAN-GP-800	91.58	93.82	89.78
Xception_Unbal	89.81	93.56	88.28
Xception_Bal_DataAug	89.93	91.50	90.56
Xception_Bal_WGAN-GP-300	90.82	92.45	90.06
Xception_Bal_WGAN-GP-500	**91.76**	**92.65**	**90.84**
Xception_Bal_WGAN-GP-800	89.05	89.99	85.64
InceptionResNetV2_Unbal	84.25	88.97	82.36
InceptionResNetV2_Bal_DataAug	86.94	90.78	85.19
IncResNetV2_Bal_WGAN-GP-300	87.02	**92.41**	**89.78**
IncResNetV2_Bal_WGAN-GP-500	86.89	91.37	89.04
IncResNetV2_Bal_WGAN-GP-800	**89.78**	91.42	88.24
InceptionV4_Unbal	89.97	94.81	90.26
InceptionV4_Bal_DataAug	90.23	95.14	93.00
InceptionV4_Bal_WGAN-GP-300	**95.36**	**96.82**	**93.69**
InceptionV4_Bal_WGAN-GP-500	95.21	96.24	92.19
InceptionV4_Bal_WGAN-GP-800	90.95	93.08	91.34

Faezehsadat Shahidi et al. [15] generated ~8000 images with 330 epochs and reached 99.49% by applying WGAN-GP. However, they synthesized high-resolution images versus low-resolution in the paper. Mai Bui Huynh Thuy and Vinh Truong Hoang [16] received 95% of accuracy to detect breast cancer on the histopathological images by StyleGAN and Pix2Pix GAN. Tasleem Kausar et al. [17] used SA-GAN to generate ~2500 images and classify them with 93% accuracy. The paper utilized the IDC dataset with patches of 198,739 non-cancerous and 78,786 cancerous images. It was generated synthetic data of cancerous class in amount of 84,000 samples and reached accuracy / sensitivity / specificity of 95.40% / 96.56% / 94.91% with VGG16 pre-trained model in 300 epochs of WGAN-GP. Additionally, we investigated four other pre-trained models with imbalance and balance datasets by data augmentation and GAN.

WGAN-GP was trained with 300, 500, and 800 epochs. The first two values of epochs yielded comparable results

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~94%-95% accuracy, which is higher than 5-10% of data augmentation with affine transformations. Thus, the results show the positive impact of the model, and the classification accuracy score improved from 84.25% to 95.40%.

VII. CONCLUSION

Author finds the field of GAN as a balance and data augmentation tool very encouraging and trustworthy to be investigated. It is proposed to continue the work and include other GANs to compare the results of generation synthetic images in future work. Besides, other pre-trained models will be trained, such as AlexNet, ResNet, MobileNet, and GoogLeNet, with different approaches to DL. The proposed system can be tested with different modalities in the medical domain, which has to be scrutinized thoroughly.

REFERENCES

[1] Gavrishchaka, Valeriy & Yang, Zhenyi & Miao, Rebecca & Senyukova, O.. (2018). Advantages of hybrid deep learning frameworks in applications with limited data. International Journal of Machine Learning and Computing. 8. 549-558. 10.18178/ijmlc.2018.8.6.744.

[2] A. Waheed, M. Goyal, D. Gupta, A. Khanna, F. Al-Turjman and P. R. Pinheiro, "CovidGAN: Data Augmentation Using Auxiliary Classifier GAN for Improved Covid-19 Detection," in IEEE Access, vol. 8, pp. 91916-91923, 2020, doi: 10.1109/ACCESS.2020.2994762.

[3] I. Sechopoulos, J. Teuwen, R. Mann, Artificial intelligence for breast cancer detection in mammography and digital breast tomosynthesis: State of the art, Seminars in Cancer Biology (2020) doi:10.1016 /j.semcancer.2020.06.002. URL https: / /dx.doi.org/10.1016/j.semcancer.2020.06.002

[4] Frid-Adar, Maayan & Klang, Eyal & Amitai, Marianne & Goldberger, Jacob & Greenspan, Heather. (2018). Synthetic data augmentation using GAN for improved liver lesion classification. 289-293. 10.1109/ISBI.2018.8363576.

[5] Uzunova, Hristina et al. "Memory-efficient GAN-based Domain Translation of High Resolution 3D Medical Images." Computerized medical imaging and graphics : the official journal of the Computerized Medical Imaging Society 86 (2020): 101801 .

[6] W. T. Tran, Computational Radiology in Breast Cancer Screening and Diagnosis Using Artificial Intelligence", Canadian Association of Radiologists Journal 72 (1) (2021) 98-108.

[7] S. M. Anwar, Medical Image Analysis using Convolutional Neural Networks: A Review", Journal of Medical Systems 42 (11) (2018) 226-226.

[8] Gong M, Chen S, Chen Q, Zeng Y, Zhang Y. Generative Adversarial Networks in Medical Image Processing. Curr Pharm Des. 2021;27(15):1856-1868. doi: 10.2174/1381612826666201125110710. PMID: 33238866.

[9] Iqbal, Talha & Ali, Hazrat. (2018). Generative Adversarial Network for Medical Images (MI-GAN). Journal of Medical Systems. 42. 10.1007/s10916-018-1072-9.

[10] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in Advances in neural information processing systems, 2014, pp. 2672– 2680

[11] X. Yi, E. Walia, P. Babyn Generative adversarial network in medical imaging: A review Medical Image Analysis, 58 (2019), Article 101552 URL:https://www.arxiv.org/abs/1809.07294

[12] R. K. Samala, H.-P. Chan, L. M. Hadjiiski, M. A. Helvie, C. Richter, K. Cha, Evolutionary pruning of transfer learned deep convolutional neural network for breast cancer diagnosis in digital breast tomosynthesis, Physics in Medicine & Biology 63 (9) (2018) 095005-095005. doi:10.1088 /1361-6560 /aabb5b.URL https:/ /dx.doi.org/ 10.1088 / 1361-6560 /aabb5b

[13] Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, and Alexei A. Efros. Image-to-Image Translation with Conditional Adversarial Networks. arXiv.org, November 2016. arXiv: 1611.07004.

[14] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks," 2017.

[15] Shahidi, Faezeh. (2021). Breast Cancer Histopathology Image Super-Resolution Using Wide-Attention GAN With Improved Wasserstein Gradient Penalty and Perceptual Loss. IEEE Access. 9. 32795-32809. 10.1109/ACCESS.2021.3057497.

[16] Thuy, Mai & Truong Hoang, Vinh. (2020). Fusing of Deep Learning, Transfer Learning and GAN for Breast Cancer Histopathological Image Classification. 10.1007/978-3-030-38364-0_23.

[17] Kausar, Tasleem & Kausar, Adeeba & Ashraf, Muhammad & Siddique, Muhammad & Wang, Mingjiang & Sajid, Muhammad & Siddique, Muhammad & Haq, Anwar & Riaz, Imran. (2021). SA-GAN: Stain Acclimation Generative Adversarial Network for Histopathology Image Analysis. Applied Sciences. 12. 288. 10.3390/app12010288.

[18] A. Das, V. K. Devarampati and M. S. Nair, "NAS-SGAN: A Semi-supervised Generative Adversarial Network Model for Atypia Scoring of Breast Cancer Histopathological Images," in IEEE Journal of Biomedical and Health Informatics, doi: 10.1109/JBHI.2021.3131103.

[19] Mariani, Giovanni & Scheidegger, Florian & Istrate, Roxana & Bekas, Costas & Malossi, A. Cristiano I.. (2018). BAGAN: Data Augmentation with Balancing GAN.

[20] Janowczyk A, Madabhushi A. Deep learning for digital pathology image analysis: A comprehensive tutorial with selected use cases. J Pathol Inform 2016;7:29

[21] Cruz-Roa, Angel & Basavanhally, Ajay & González, Fabio & Gilmore, Hannah & Feldman, Michael & Ganesan, Shridar & Shih, Natalie & Tomaszewski, John & Madabhushi, Anant. (2014). Automatic detection of invasive ductal carcinoma in whole slide images with Convolutional Neural Networks. Progress in Biomedical Optics and Imaging - Proceedings of SPIE. 9041. 10.1117/12.2043872.

[22] M. Arjovsky, S. Chintala, and L. Bottou. Wasserstein gan. arXiv preprint arXiv:1701.07875, 2017

[23] Gulrajani, Ishaan & Ahmed, Faruk & Arjovsky, Martin & Dumoulin, Vincent & Courville, Aaron. (2017). Improved Training of Wasserstein GANs.

[24] Bhatia, Snehal and Rozenn Dahyot. "Using WGAN for Improving Imbalanced Classification Performance." AICS (2019).

[25] Geras KJ, Mann RM, Moy L. Artificial Intelligence for Mammography and Digital Breast Tomosynthesis: Current Concepts and Future Perspectives. Radiology. 2019 Nov;293(2):246-259. doi: 10.1148/radiol.2019182627. Epub 2019 Sep 24. PMID: 31549948; PMCID: PMC6822772.

[26] He, S., Wu, Q. H. and Saunders, J. R. (2009) 'Breast cancer diagnosis using an artificial neural network trained by group search optimizer', Transactions of the Institute of Measurement and Control, 31(6), pp. 517–531. doi: 10.1177/0142331208094239.

[27] Ahmed, Luqman & Iqbal, Muhammad & Aldabbas, Hamza & Saeed, Saqib. (2020). Images data practices for Semantic Segmentation of Breast Cancer using Deep Neural Network. Journal of Ambient Intelligence and Humanized Computing. 10.1007/s12652-020-01680-1.

[28] McKinney, Scott & Sieniek, Marcin & Godbole, Varun & Godwin, Jonathan & Antropova, Natasha & Ashrafian, Hutan & Back, Trevor & Chesus, Mary & Corrado, Greg & Darzi, Ara & Etemadi, Mozziyar & Garcia-Vicente, Florencia & Gilbert, Fiona & Halling-Brown, Mark & Hassabis, Demis & Jansen, Sunny & Karthikesalingam, Alan & Kelly, Christopher & King, Dominic & Shetty, Shravya. (2020). International evaluation of an AI system for breast cancer screening. Nature. 577. 89-94. 10.1038/s41586-019-1799-6.

[29] Ayer, Turgay & Chen, Qiushi & Burnside, Elizabeth. (2013). Artificial Neural Networks in Mammography Interpretation and Diagnostic Decision Making. Computational and mathematical methods in medicine. 2013. 832509. 10.1155/2013/832509.

[30] Zebari, Dilovan & Haron, Habibollah & Zeebaree, Subhi & Zeebaree, Diyar. (2019). Enhance the Mammogram Images for Both Segmentation and Feature Extraction Using Wavelet Transform. 100-105. 10.1109/ICOASE.2019.8723779.)

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